University of the Witwatersrand
School of Computer Science and Applied Mathematics
COMS4070A & COMS7081A:
Computational Neuroscience
Population Coding, Dynamics and Memory Tutorial — Memo
(Lectures 4–6)

How to use this tutorial

This tutorial combines **single best answer and written questions** (concepts and derivations) with a **companion notebook** (`tutorial_notebook_solutions.ipynb`). These notebook tasks are entirely optional extension material and will not be in the test! The aim is to help build intuition for derivations you do on paper, so I have filled in the programming answers already. Where a question is marked **[Notebook Task X]**, do the written part first and then use the notebook to confirm your answer.

The written questions give an example of the types of questions you can expect in a test or exam. Some may be repetitive, to help calibrate to the different ways the same concept may be asked. Marks are an indication only.

1 Warm up: Quiz style single best answer questions

1. **[Multiple Choice]** For a Poisson spike count with rate λ, the variance of the count is:

   a) λ b) λ^2 c) $\sqrt{\lambda}$ d) 1

   Memo: (a).

2. **[True/False]** The maximum likelihood estimate for a linear neuron with Gaussian noise of constant variance is the least-squares solution.

   a) True b) False

   Memo: True.

3. **[True/False]** The spike-triggered average recovers a neuron's linear filter for any stimulus distribution.

   a) True b) False

   Memo: False: it needs e.g. Gaussian white noise; correlated stimuli bias it.

4. **[Multiple Choice]** In the GLM $r_t \sim \text{Poisson}(f(\mathbf{s}_t \cdot \mathbf{w}))$, the main role of the non-linearity f is to:

   a) reduce the dimensionality of the stimulus.
   b) map the linear drive to a non-negative firing rate.
   c) add noise to the response.
   d) make the log-likelihood linear in $\mathbf{w}$.

   Memo: (b).

5. **[Multiple Choice]** For $\tau\dot{r} = -r + wr + I$ with $w = 0.75$, the effective time constant is:

   a) $\tau/4$ b) 0.75τ c) 4τ d) undefined (unstable)

   Memo: (c): $\tau/(1-w) = 4\tau$.

6. **[Multiple Choice]** A fixed point has Jacobian eigenvalues $\lambda = -0.5 \pm 2i$. Nearby trajectories:

   a) grow without oscillating.
   b) decay without oscillating.
   c) spiral inwards (damped oscillations).
   d) form a stable limit cycle.

   Memo: (c).

7. **[True/False]** In a balanced network, excitatory and inhibitory inputs are each individually small.

   a) True b) False

   Memo: False: each is large ($\sim \sqrt{K}$); it is their sum that is small.

8. **[True/False]** In an inhibition-stabilised network, increasing external input to the inhibitory population can decrease inhibitory firing at steady state.

   a) True b) False

   Memo: True (paradoxical effect).

9. **[Multiple Choice]** For random dense patterns in a classical Hopfield network with N neurons, the maximum number of patterns that can be reliably stored is approximately:

   a) 0.138 b) $0.138N$ c) $0.138N^2$ d) 2^N

   Memo: (b).

10. **[Matching Pairs]** Match each variable to the attractor geometry best suited to represent it: (i) horizontal eye position, (ii) head direction, (iii) one of several stored odours.

    a) point attractors b) line attractor c) ring attractor

    Memo: (i)–b, line, (ii)–c, ring, (iii)–a, point.

2 Neural coding and population representations (Lecture 4)

2.1 Tuning curves and the spike-triggered average

A.1 Define a *tuning curve*. Name four properties of a neuron's code that can be read off its tuning curve. [4 Marks]

Memo: A tuning curve is the mean response (e.g. firing rate) of a neuron as a function of a stimulus variable, $f(s) = \mathbb{E}[r \mid s]$. You could also think of it as the single-neuron representation of the $S \to X$ map. From the tuning curve, we can read:

- The preferred stimulus (location of the peak)
- The tuning width (selectivity)
- The baseline (response to non-preferred or no stimulus)
- The gain (peak height above baseline).

A.2 Explain how the spike-triggered average (STA) is computed. Why do stimulus features that are unrelated to spiking "average away"? What assumption about the stimulus makes the STA recover the neuron's linear filter? [4 Marks]

Memo: For each spike at time t_k, take the stimulus snippet $s(t_k - \tau)$ preceding it, and average over all N spikes: $\mathrm{STA}(\tau) = \frac{1}{N}\sum_k s(t_k - \tau)$. Components of the stimulus not consistently present before spikes are equally likely to be positive or negative, so they cancel when averaged, and only the feature that reliably drives spikes survives. For Gaussian white-noise (more generally, spherically symmetric) stimuli the STA is proportional to the linear filter (this is from something called Bussgang's Theorem: you do *not* need to know this theorem). With correlated stimuli it is biased by the stimulus correlations, so we must use $(S^\top S)^{-1} S^\top \mathbf{r}$ instead.

A.3 Distinguish *encoding* from *decoding*. In which sense is the STA used for each? [3 Marks]

Memo: Encoding models the (neural!) response given the stimulus, $P(r \mid s)$; decoding infers the stimulus (or feature) from the response, $P(s \mid r)$. Averaging stimuli conditioned on spikes is a decoding view: which stimulus preceded a spike? Using the STA as a linear filter to predict $\hat{r}(t) = \int s(t - \tau) w(\tau)\, d\tau$ is an encoding model.

A.4 Why is the Poisson distribution a natural first choice for spike counts in a time window? State one way in which real neurons deviate from it. [3 Marks]

Memo: Spike counts are non-negative integers. If spikes are rare, independent events at a constant rate, the count in a window is Poisson. Additionally (for interest), its variance equals its mean, which is roughly what is seen in cortex from data. You could choose any reasonable deviation. The most obvious is that refractoriness makes spike trains more "regular" than Poisson at short time scales (i.e. one cannot have infinitely many spikes in one time bin / consecutive points in time are not really independent for real neurons as Poisson assumes — recall the potassium correction channels for the refractory hyperpolarisation after a spike). I would also accept: bursting, adaptation or shared input that change the variance=mean fact.

2.2 From a neuron with a constant rate to MLE

A neuron fires at an unknown constant rate λ. We count its spikes $n_1, \ldots, n_n$ in n independent 1 s windows, $n_i \sim \text{Poisson}(\lambda)$, and write $S = \sum_i n_i$.

A.5 **[Notebook Task A1 (extension)]** Spike counts are often modelled as Gaussian.

(a) Derive the MLEs $\hat{\mu}$ and $\hat{\sigma}^2$ for i.i.d. Gaussian data, following the lecture's approach for the Poisson MLE if you get stuck. [4 Marks]

(b) If the counts are actually Poisson, what values for mean and variance do you expect the counts to approach? [1 Mark]

(c) Given that the two models fit high-rate data about equally well, why might we still prefer the Poisson model? (Bonus points if you know *why* they both fit high rates well, but this is not expected!) [2 Marks]

Memo: (a) $\log P = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (n_i - \mu)^2$. $\partial_\mu = 0 \Rightarrow \hat{\mu} = \bar{n}$; $\partial_{\sigma^2} = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n}\sum_i (n_i - \bar{n})^2$ (note $1/n$, not $1/(n-1)$). (b) For Poisson data both approach λ (mean = variance). (c) The Gaussian has two free parameters to the Poisson's one, and with extra flexibility it can only fit as well or better. We should prefer the simpler model that explains the data equally well (Occam's razor), e.g. via BIC $= k \ln n - 2\log \hat{L}$ or, fully Bayesian, the evidence $p(\mathcal{D} \mid \mathcal{M})$. **Extension info (not test-able):** At low rates counts are discrete, non-negative and right-skewed. A symmetric Gaussian puts probability on negative counts and misses the skew, so its likelihood is clearly lower (notebook: $\Delta \log L \approx 9$ at $\lambda = 2$). At high rates $\text{Poisson}(\lambda) \approx \mathcal{N}(\lambda, \lambda)$ (central limit theorem), so the maximised likelihoods are almost equal.

Now suppose a neuron responds to contrast $s_i \geq 0$ on trial i ($i = 1, \ldots, n$) with a linear rate $\lambda_i = \theta s_i$.

A.6 **[Notebook Task A2–A3]** If the spike count is $n_i \sim \text{Poisson}(\theta s_i)$, write the likelihood $P(\mathbf{n} \mid \mathbf{s}, \theta)$ and state the assumption needed to write it as a product over trials. Derive the maximum likelihood estimate $\hat{\theta}_{\text{ML}}$ and verify it is a maximum. [6 Marks]

Memo: Assuming trials are conditionally independent given the stimulus,

$$\log P(\mathbf{n} \mid \mathbf{s}, \theta) = \sum_i \left[n_i \log(\theta s_i) - \theta s_i - \log n_i! \right].$$

Setting $\frac{d}{d\theta} = \sum_i \frac{n_i}{\theta} - \sum_i s_i = 0$ gives $\boxed{\hat{\theta}_P = \sum_i n_i / \sum_i s_i}$. The second derivative $-\sum_i n_i/\theta^2 < 0$ confirms it is a maximum.

2.3 Generalised linear models

A.7 Consider the linear-Gaussian GLM $\mathbf{r} = S\mathbf{w} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I)$. Derive $\hat{\mathbf{w}} = (S^\top S)^{-1} S^\top \mathbf{r}$. When does this quantity fail to exist? [4 Marks]

Memo: $\log P = -\frac{1}{2\sigma^2}\|\mathbf{r} - S\mathbf{w}\|^2 + \text{const}$. Gradient: $\frac{1}{\sigma^2} S^\top(\mathbf{r} - S\mathbf{w}) = 0 \Rightarrow S^\top S\mathbf{w} = S^\top \mathbf{r}$. This requires $S^\top S$ to be invertible, which fails if the columns of S are linearly dependent (e.g. more filter coefficients than time points, or a stimulus that doesn't vary enough). Regularisation (ridge) fixes this.

A.8 **[Notebook Task A4]** For the Poisson GLM $r_t \sim \text{Poisson}(\Delta \lambda_t)$ with $\lambda_t = \exp(\mathbf{s}_t \cdot \mathbf{w})$: [7 Marks]

(a) Write down the log-likelihood (dropping terms that do not depend on $\mathbf{w}$).

(b) Compute its gradient with respect to $\mathbf{w}$.

Memo: (a) $\mathcal{L}(\mathbf{w}) = \sum_t [r_t\, \mathbf{s}_t \cdot \mathbf{w} - \Delta e^{\mathbf{s}_t \cdot \mathbf{w}}] = \mathbf{r}^\top S\mathbf{w} - \Delta \mathbf{1}^\top \exp(S\mathbf{w})$. (b) $\nabla_{\mathbf{w}} \mathcal{L} = S^\top(\mathbf{r} - \Delta \exp(S\mathbf{w}))$: stimulus-weighted prediction error, the same form as in linear regression.

2.4 Populations and latent variables

A.9 What is a population code? Give two reasons why a population of noisy neurons with different tuning curves can represent a stimulus better than any single neuron. [4 Marks]

Memo: In a population code, a stimulus is assumed to be represented by the pattern of activity $\boldsymbol{r} = (r_1, \ldots, r_N)$ across many neurons with different tuning curves $f_i(s)$. Choose any two. (i) First, it can reduce noise. Independent noise averages out, so decoding error falls. (ii) Diverse tuning covers the whole stimulus range, including places where one neuron's tuning curve is flat and uninformative. (iii) The population pattern can also represent uncertainty (a distribution), not just a point estimate.

A.10 In PCA we write $X \approx ZW^\top$ with $X \in \mathbb{R}^{T \times N}$ and $Z \in \mathbb{R}^{T \times K}$, $K \ll N$. Explain what Z and W represent in terms of neural activity. Give an example of a latent variable that the experimenter does not control but which might dominate the population activity. [4 Marks]

Memo: Each row of Z gives the low-dimensional latent state at one time point. Each column of W is a direction in neuron space: how each latent dimension is expressed across the N neurons. So $\boldsymbol{r}_t \approx W\boldsymbol{z}_t$. PCA picks the directions of maximal variance. Examples of uncontrolled latents: arousal, attention, running speed or other movements, internal beliefs, task strategy.

3 Recurrent dynamics and E–I balance (Lecture 5)

3.1 From spiking networks to rate models

B.1 A network of N conductance-based neurons has $2N$ state variables and products like $g_j(t)V_i(t)$. Explain what we gain and what we lose by replacing it with the rate model $\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$. [4 Marks]

Memo: Gain: far fewer variables, a model that can be analysed (fixed points, linearisation, phase planes), and parameters (W, ϕ, $\boldsymbol{I}$) that map onto circuit-level questions. Lose: spike timing and synchrony, single-neuron biophysics (channels, refractoriness), synaptic time courses, and fluctuations caused by spiking. The model assumes activity can be temporally averaged into a rate. Other reasonable answers also accepted.

B.2 Explain the role of each term in $\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$: τ, $-\boldsymbol{r}$, $W\boldsymbol{r}$, $\boldsymbol{I}$ and ϕ. [5 Marks]

Memo: τ: time scale of the population response. $-\boldsymbol{r}$: leak, decay towards zero without input. $W\boldsymbol{r}$: recurrent input from other neurons in the network (sign gives E or I). $\boldsymbol{I}$: external (feedforward) input. ϕ: input–output non-linearity (e.g. F–I curve), converting total input into a rate.

3.2 A single recurrent population

Take $\phi(x) = x$ and a single population: $\tau \frac{dr}{dt} = -r + wr + I$.

B.3 Derive the fixed point r^* and the effective time constant τ_{eff} of small perturbations about it. Complete the table below and describe in words what recurrent excitation does to the network's gain and speed. [6 Marks]

w	gain r^*/I	τ_{eff}/τ
0		
0.5		
0.9		
0.99		

Memo: $0 = (w-1)r^* + I \Rightarrow r^* = I/(1-w)$. With $r = r^* + \delta r$: $\tau \dot{\delta r} = (w-1)\delta r \Rightarrow \delta r(t) = \delta r(0) e^{-(1-w)t/\tau}$, so $\tau_{\text{eff}} = \tau/(1-w)$. Both gain and τ_{eff}/τ equal $1/(1-w)$: $1, 2, 10, 100$. Recurrent excitation amplifies input and extends the time constant: the network becomes more sensitive and integrates over a longer window.

B.4 What happens when $w > 1$? When $w < 0$? What stops the activity of real networks from growing without bound? [4 Marks]

Memo: $w > 1$: the fixed point $I/(1-w)$ still exists but is unstable, since perturbations grow as $e^{(w-1)t/\tau}$: runaway excitation. $w < 0$ (self-inhibition): gain < 1 and $\tau_{\text{eff}} < \tau$, so the response is suppressed and faster. In real networks, saturating non-linearities ϕ, inhibitory feedback and adaptation bound the activity.

B.5 What happens at exactly $w = 1$? Why might this be useful, and why is it biologically delicate? [3 Marks]

Memo: $\tau \dot{r} = I$: the population integrates its input, and with $I = 0$ every r is a fixed point, which is a line attractor. This could hold a memory of accumulated input (e.g. the oculomotor integrator holding eye position). It requires fine-tuning: any w slightly away from 1 makes the memory decay ($w < 1$) or blow up ($w > 1$).

3.3 The two-population E–I network

$$\tau_{\text{E}} \dot{r}_{\text{E}} = -r_{\text{E}} + \phi(w_{\text{EE}} r_{\text{E}} - w_{\text{EI}} r_{\text{I}} + I_{\text{E}}), \qquad \tau_{\text{I}} \dot{r}_{\text{I}} = -r_{\text{I}} + \phi(w_{\text{IE}} r_{\text{E}} - w_{\text{II}} r_{\text{I}} + I_{\text{I}})$$

with all $w \geq 0$. For B.8–B.12, take $\phi(x) = x$, $\tau_{\text{E}} = \tau_{\text{I}} = 1$, and

$$w_{\text{EE}} = 2, \quad w_{\text{EI}} = 2, \quad w_{\text{IE}} = 2, \quad w_{\text{II}} = 0.5, \quad I_{\text{E}} = 1, \quad I_{\text{I}} = 0.5.$$

B.6 Draw the E–I circuit, labelling each connection with its sign. Describe the feedback loop that makes inhibition stabilising. [3 Marks]

Memo: E→E (+), E→I (+), I→E (−), I→I (−). If E activity rises, it drives I, which in turn suppresses E: $E \uparrow \rightarrow I \uparrow \rightarrow E \downarrow$ (negative feedback).

B.7 Define a *nullcline*. Why must every fixed point lie on the intersection of the two nullclines? [2 Marks]

Memo: The r_{E}-nullcline is the set of states where $\dot{r}_{\text{E}} = 0$ (likewise for r_{I}). A fixed point needs $\dot{r}_{\text{E}} = \dot{r}_{\text{I}} = 0$ simultaneously, so it lies on both.

B.8 **[Notebook Task B1]** Derive both nullclines as functions $r_\mathrm{I}(r_\mathrm{E})$ and find the fixed point. [5 Marks]

Memo: $\dot{r}_\mathrm{E} = 0$: $0 = (2-1)r_\mathrm{E} - 2r_\mathrm{I} + 1 \Rightarrow r_\mathrm{I} = \frac{1}{2}(r_\mathrm{E}+1)$. $\dot{r}_\mathrm{I} = 0$: $0 = 2r_\mathrm{E} - 1.5r_\mathrm{I} + 0.5 \Rightarrow r_\mathrm{I} = \frac{1}{3}(4r_\mathrm{E}+1)$. Equate: $3r_\mathrm{E} + 3 = 8r_\mathrm{E} + 2 \Rightarrow \boxed{r_\mathrm{E}^* = 0.2,\ r_\mathrm{I}^* = 0.6}$.

B.9 Write down the Jacobian J of this system. [4 Marks]

Memo: $J = \begin{pmatrix} w_{\mathrm{EE}} - 1 & -w_{\mathrm{EI}} \\ w_{\mathrm{IE}} & -(1 + w_{\mathrm{II}}) \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 2 & -1.5 \end{pmatrix}$. [Extension, not in test]: Note that, for a linear system, does J not depend on which fixed point we linearise around. For $\dot{\boldsymbol{r}} = A\boldsymbol{r} + \boldsymbol{I}$ the derivatives are constant, so $J = A$ everywhere; the perturbation dynamics are exactly the full dynamics.

B.10 **[Notebook Task B2]** The eigenvalues λ of J are given by analysing the determinant of the Jacobian, i.e. since $\mathrm{tr}J = -0.5$, $\det J = -1.5 + 4 = 2.5$ and $\lambda^2 + 0.5\lambda + 2.5 = 0 \Rightarrow \lambda = -0.25 \pm i\sqrt{2.4375} \approx -0.25 \pm 1.56i$. Classify the fixed point (refer to Figure 2). From this, you can also predict the oscillation period and the decay time constant of perturbations, and check the prediction against Figure 1 (not expected for the test but included for interest). [3 Marks]

Memo: Negative real part, non-zero imaginary part: a *stable spiral* (damped oscillations). Period $2\pi/\omega \approx 4.0\,\tau$; amplitude decays as $e^{-0.25t}$, i.e. time constant 4τ. So the amplitude falls by roughly e^{-1} per cycle, consistent with the spiralling trajectories in the figure.

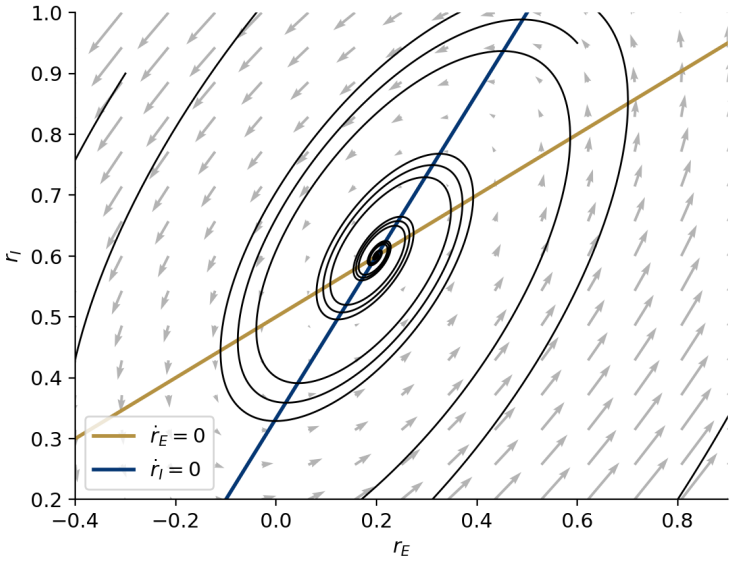


Figure 1: Phase plane of the linear E–I network in B.6–B.10: nullclines, vector field (grey) and three trajectories (black).

B.11 **Inhibition-stabilised networks.** [8 Marks] **[Notebook Task B3]**

(a) Explain in words why the excitatory population on its own (with r_I held fixed) would be unstable.

(b) Solve for the fixed point r_I^* as a function of I_I (keeping $I_\mathrm{E} = 1$). What happens to inhibitory activity when we give the inhibitory population *more* input? Explain this "paradoxical effect" intuitively.

(c) Repeat (b) with $w_{EE} = 0.5$. What changes?

Memo: (a) With r_I fixed, $\dot{\delta r}_E = (w_{EE} - 1)\delta r_E = +\delta r_E$: perturbations grow. The network is only stable because of inhibitory feedback: an inhibition-stabilised network (ISN). (b) From the E-nullcline, $r_E = 2r_I - 1$. Substituting into $\dot{r}_I = 0$: $2(2r_I - 1) - 1.5r_I + I_I = 0 \Rightarrow r_I^* = (2 - I_I)/2.5 = 0.8 - 0.4I_I$ (and $r_E^* = 0.6 - 0.8I_I$). Driving I *lowers* steady-state inhibition. Intuition: extra drive to I suppresses E. Because E is strongly self-excitatory, E activity drops a lot, and the resulting loss of excitatory drive to I outweighs the extra external input. Equivalently, at steady state r_I must lie on the E-nullcline, which has positive slope when $w_{EE} > 1$, so r_I moves in the same direction as r_E. (c) With $w_{EE} = 0.5$: $r_E = 2 - 4r_I$ and $r_I^* = (4 + I_I)/9.5$, which increases with I_I. The E-nullcline now has negative slope: no paradoxical effect. The paradoxical response is an experimental signature of an ISN.

3.4 Non-linear E–I dynamics and balanced networks

B.12 Restore the non-linearity ϕ. Derive all four entries of the Jacobian of the Wilson–Cowan system at a fixed point (r_E^*, r_I^*). [4 Marks]

Memo: Let $\phi'_E = \phi'(w_{EE}r_E^* - w_{EI}r_I^* + I_E)$ and $\phi'_I = \phi'(w_{IE}r_E^* - w_{II}r_I^* + I_I)$. By the chain rule,

$$J = \begin{pmatrix} \dfrac{-1 + w_{EE}\phi'_E}{\tau_E} & \dfrac{-w_{EI}\phi'_E}{\tau_E} \\ \dfrac{w_{IE}\phi'_I}{\tau_I} & \dfrac{-1 - w_{II}\phi'_I}{\tau_I} \end{pmatrix}.$$

The effective connection strengths are $w\phi'$: the weights scaled by the local slope (gain) of the input–output function.

B.13 Give one example of how a disturbance of E–I balance has been linked to a clinical condition, and explain it in dynamical terms. [3 Marks]

Memo: E.g. epilepsy: too little effective inhibition (or too much recurrent excitation) pushes the network past a stability boundary, into runaway or oscillatory activity (seizures). Other examples from the lecture: altered E/I ratio in autism or schizophrenia models (Jardri & Denève, 2013; Gao et al., 2017). Credit any answer that links the change to gain, stability or eigenvalues.

4 Attractors, memory and canonical computation (Lecture 6)

4.1 Multiple stable states

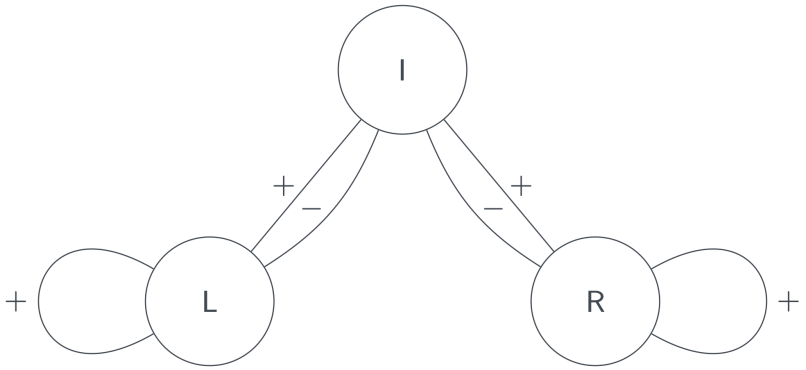
C.1 Using a sketch of $y = r$ and $y = \phi(wr + I)$ for a single population, explain why a linear ϕ gives at most one fixed point but a sigmoidal ϕ can give three. [3 Marks]

Memo: Fixed points satisfy $r^* = \phi(wr^* + I)$: intersections of the diagonal with the curve. Two straight lines cross at most once (unless they coincide). A sigmoid that is flat, then steeper than the diagonal, then flat again can cross it three times.

C.2 Define an *attractor* and a *basin of attraction*. Distinguish point attractors from continuous attractors and give a neural example of each of a point, a line and a ring attractor. [6 Marks]

Memo: Attractor: a state (or set of states) towards which nearby trajectories converge. Basin: the set of initial states that converge to a given attractor. Point attractor: isolated stable fixed point (a discrete memory; a decision in a winner-take-all network). Continuous attractor: a connected manifold of marginally stable states along which the state can sit anywhere. Line attractor: oculomotor integrator holding eye position. Ring attractor: head-direction cells, where the activity bump position encodes heading.

C.3 State the three ingredients a recurrent circuit needs to implement winner-take-all decisions, and explain how the circuit shown below (from the Lecture 6 WTA slide) provides each. [3 Marks]



$$E_L \uparrow\!\rightarrow I \uparrow\!\rightarrow E_R \downarrow$$

One population wins; the other is suppressed.

Memo: Positive feedback (recurrent self-excitation of L and R), stabilisation (shared inhibitory population prevents runaway), competition (L drives I which suppresses R, and vice versa).

4.2 Hopfield networks

Assume that neurons are binary, $s_i \in \{-1, +1\}$. We store P patterns ξ^μ with the Hebbian rule

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu ,$$

$J_{ii} = 0$ and update $s_i \leftarrow \text{sign}(h_i)$ with local field $h_i = \sum_j J_{ij} s_j$.

C.4 Interpret the Hebbian rule: for which pairs of neurons does a single pattern make J_{ij} positive, and for which negative? Why is J symmetric? [3 Marks]

Memo: Pattern μ adds $+1/N$ when $\xi_i^\mu = \xi_j^\mu$ (both $+1$ *or* both -1) and $-1/N$ when they disagree. So, with ± 1 units, neurons that are "off together" are also wired together. $J_{ij} = J_{ji}$ because $\xi_i \xi_j = \xi_j \xi_i$.

C.5 Why do we set $J_{ii} = 0$? [1 Mark]

Memo: $J_{ii} = P/N$ would be a constant self-excitation that tends to freeze every neuron in its current state (every state becomes more stable, including wrong ones). Removing it also simplifies the energy argument.

C.6 **[Notebook Task C1] By hand.** Store the two patterns ($N = 6$)

$$\xi^1 = (+1, +1, +1, -1, -1, -1), \qquad \xi^2 = (+1, +1, -1, +1, -1, -1).$$

(a) Compute the 6×6 weight matrix J (you may give $6J$).

(b) The network starts in the cue $\boldsymbol{s} = (+1, +1, +1, -1, +1, -1)$. Compute the overlap $m^\mu = \frac{1}{N}\sum_i \xi_i^\mu s_i$ for each of the two patterns.

(c) Compute all six local fields and the state after one synchronous update. Which pattern has been retrieved?

(d) Compute the energy $E(\boldsymbol{s}) = -\frac{1}{2}\sum_{i,j} J_{ij} s_i s_j$ of the cue and of the retrieved state.

[10 Marks]

Memo: (a)

$$6J = \begin{pmatrix} 0 & 2 & 0 & 0 & -2 & -2 \\ 2 & 0 & 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 \\ -2 & -2 & 0 & 0 & 0 & 2 \\ -2 & -2 & 0 & 0 & 2 & 0 \end{pmatrix}$$

(entries are $+2$ where the patterns agree on both pairs, 0 where one pattern says "same" and the other "different"). (b) The cue is ξ^1 with neuron 5 flipped: $m^1 = 4/6 = 2/3$, $m^2 = 0$. (c) $6\boldsymbol{h} = (2, 2, 2, -2, -6, -2)$, so $\mathrm{sign}(\boldsymbol{h}) = (+, +, +, -, -, -) = \xi^1$. The corrupted neuron receives the strongest corrective field and flips back; ξ^1 is retrieved ($m^1 = 1$, $m^2 = 1/3$). (d) $E(\text{cue}) = -1/3$; $E(\xi^1) = -\frac{1}{2}\left[\frac{1}{N}\sum_\mu (\xi^\mu \cdot \xi^1)^2 - P\right] = -\frac{1}{2}\left[\frac{36+4}{6} - 2\right] = -7/3$. Retrieval lowered the energy.

C.7 Now start the network exactly in ξ^1. Compute the local fields. Which neurons are held in place least strongly, and why? Relate your answer to the overlap $\xi^1 \cdot \xi^2$. [4 Marks]

Memo: $6\boldsymbol{h} = (6, 6, 2, -2, -6, -6)$: all signs match ξ^1 (it is stable), but neurons 3 and 4 have field only $\pm 1/3$ instead of ± 1. These are exactly the neurons where ξ^1 and ξ^2 disagree. Because $\xi^1 \cdot \xi^2 = 2 \neq 0$, pattern 2 contributes crosstalk: $+\xi_i^1/6$ where the patterns agree, $-\xi_i^1/2$ where they disagree. Non-orthogonal memories weaken each other on the bits where they differ.

C.8 Suppose the network is exactly in memory ξ^μ. Substitute the Hebbian weights into h_i and separate the result into a *signal* term and a *crosstalk* term. Under what condition on the crosstalk is ξ^μ a fixed point? [5 Marks]

Memo: $h_i = \frac{1}{N}\sum_{j\neq i}\sum_\nu \xi_i^\nu \xi_j^\nu \xi_j^\mu = \underbrace{\frac{N-1}{N}\xi_i^\mu}_{\text{signal}} + \underbrace{\sum_{\nu\neq\mu} \xi_i^\nu \frac{1}{N}\sum_{j\neq i} \xi_j^\nu \xi_j^\mu}_{\text{crosstalk } C_i^\mu}$. The signal term uses $\xi_j^\mu \xi_j^\mu = 1$. ξ^μ is a fixed point iff $\xi_i^\mu h_i > 0$ for all i, i.e. the crosstalk never opposes the signal by more than ≈ 1: $\xi_i^\mu C_i^\mu > -\frac{N-1}{N}$. For random uncorrelated patterns the overlaps are ≈ 0 in the large-N limit, so $h_i \approx \xi_i^\mu$.

C.9 Show that if the stored patterns are mutually orthogonal ($\xi^\mu \cdot \xi^\nu = 0$ for $\mu \neq \nu$) every pattern is a fixed point for any $P < N$. Relate this to the idea of *whitening* the representation, and explain why storing many similar memories (e.g. similar faces) is hard. [5 Marks]

Memo: With orthogonal patterns, $\sum_{j\neq i}\xi_j^\nu\xi_j^\mu = 0 - \xi_i^\nu\xi_i^\mu$, so each other pattern contributes $-\xi_i^\mu/N$ and $h_i = \xi_i^\mu\frac{N-1-(P-1)}{N} = \xi_i^\mu\frac{N-P}{N}$, which has the correct sign for all $P < N$. Whitening/decorrelating the representation reduces pattern overlaps, hence crosstalk, hence raises capacity. Similar memories have large overlaps, so their crosstalk is large and they interfere or merge.

C.10 List three biologically unrealistic features of the classical Hopfield network and, for each, a modification that addresses it. [6 Marks]

Memo: Any three, e.g.: all-to-all connectivity → sparse/diluted connectivity; binary units and discrete time → continuous rate units (Hopfield, 1984); dense ±1 patterns and low capacity → sparse coding with thresholds, which raises capacity; symmetric weights → asymmetric (the energy function is lost but retrieval often survives); a single neuron has both positive and negative outgoing weights, violating Dale's law → separate E and I populations; no "no memory" state → add a quiescent background attractor; one-shot Hebbian storage → online plasticity rules.

5 List of Symbols and Equations

5.1 Neural coding (Lecture 4)

Symbol	Definition
s, $\mathbf{s}_t$	Stimulus (scalar / stimulus history vector at time t)
S	Design matrix of (shifted) stimuli
r, n	Response: rate or spike count
λ	Firing rate (Poisson mean)
θ, $\mathbf{w}$	Model parameters / linear filter
$f(\cdot)$	GLM non-linearity
Δ	Time bin width
$\boldsymbol{r} = (r_1, \ldots, r_N)$	Population response
$\boldsymbol{z}$, W	Latent variables and their mapping to neurons

Spike-triggered average: $\mathrm{STA}(\tau) = \dfrac{1}{N}\sum_{k=1}^{N} s(t_k - \tau)$

Maximum likelihood: $\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} \prod_i P(r_i \mid s_i, \theta)$

Poisson distribution: $P(n \mid \lambda) = \dfrac{\lambda^n e^{-\lambda}}{n!}$

Linear-Gaussian GLM: $\hat{\mathbf{w}} = (S^\top S)^{-1} S^\top \mathbf{r}$

Poisson GLM log-likelihood: $\mathcal{L} = \mathbf{r}^\top \log f(S\mathbf{w}) - \Delta \mathbf{1}^\top f(S\mathbf{w})$

Bernoulli GLM log-likelihood: $\mathcal{L} = \mathbf{r}^\top \log f(S\mathbf{w}) + (\mathbf{1} - \mathbf{r})^\top \log[1 - f(S\mathbf{w})]$

PCA / linear latent model: $\boldsymbol{r}_t = W\boldsymbol{z}_t + \boldsymbol{\epsilon}_t$

5.2 Recurrent dynamics (Lecture 5)

Symbol	Definition
τ, τ_{E}, τ_{I}	Population time constants
W, w_{ab}	Recurrent weights (from population b to a)
ϕ, ϕ'	Input–output non-linearity and its slope (gain)
$\boldsymbol{r}^*$	Fixed point
J	Jacobian at a fixed point
$\lambda = \alpha \pm i\omega$	Eigenvalues of J (decay rate, angular frequency)
K	Number of inputs per neuron

Rate model: $\tau\dot{\boldsymbol{r}} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$

Single linear population: $r^* = \dfrac{I}{1-w}, \quad \tau_{\text{eff}} = \dfrac{\tau}{1-w}$

Wilson–Cowan Jacobian:

$$J = \begin{pmatrix} \dfrac{-1 + w_{\text{EE}}\phi'_{\text{E}}}{\tau_{\text{E}}} & \dfrac{-w_{\text{EI}}\phi'_{\text{E}}}{\tau_{\text{E}}} \\ \dfrac{w_{\text{IE}}\phi'_{\text{I}}}{\tau_{\text{I}}} & \dfrac{-1 - w_{\text{II}}\phi'_{\text{I}}}{\tau_{\text{I}}} \end{pmatrix}$$

2×2 **stability:** stable iff $\text{Tr}(J) < 0$ and $\det J > 0$; oscillatory iff $(\text{Tr}(J))^2 < 4 \det J$.

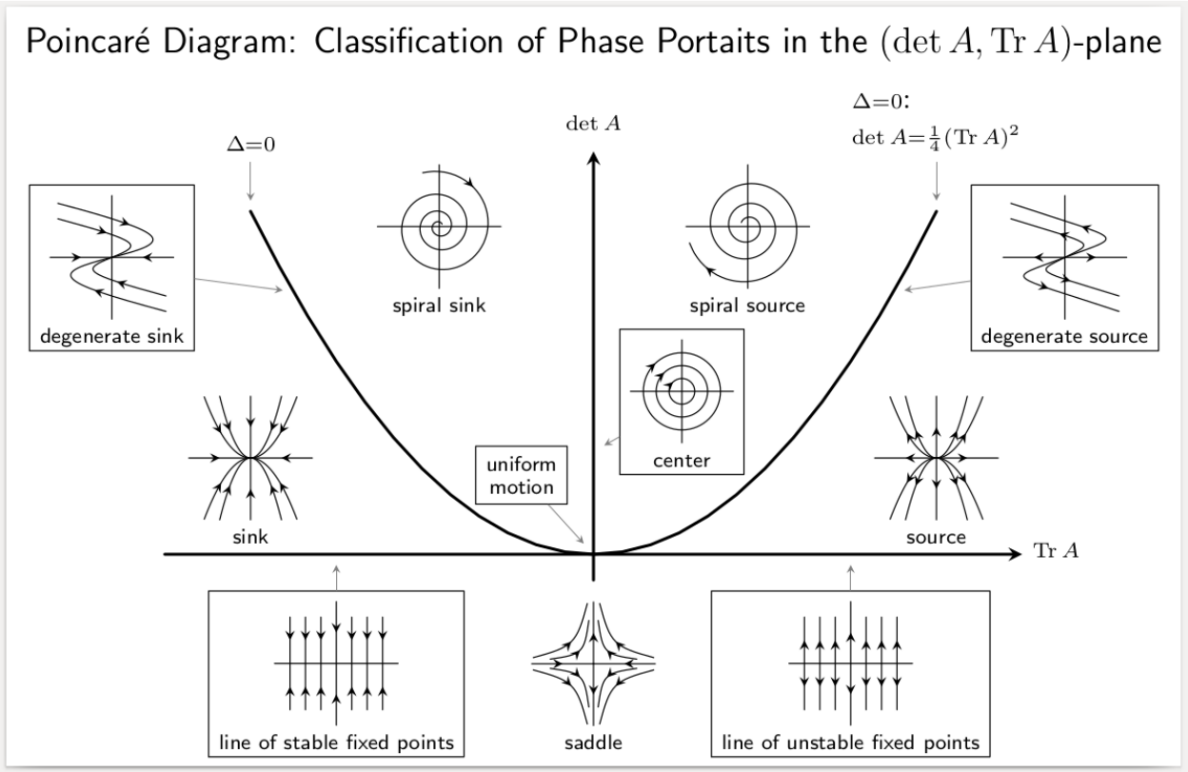


Figure 2: Poincaré diagram to aid classification of fixed points in 2D phase planes.

5.3 Attractors and Hopfield networks (Lecture 6)

Symbol	Definition
$s_i \in \{-1, +1\}$	State of neuron i
ξ^μ	Stored pattern μ ($\mu = 1, \ldots, P$)
J_{ij}	Hopfield weight from j to i
h_i	Local field (total recurrent input)
m^μ	Overlap with pattern μ
$\alpha = P/N$	Memory load

Hebbian storage: $J_{ij} = \dfrac{1}{N} \displaystyle\sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu, \quad J_{ii} = 0$

Update rule: $s_i \leftarrow \operatorname{sign}\Big(\displaystyle\sum_j J_{ij} s_j \Big)$

Overlap: $m^\mu = \dfrac{1}{N} \displaystyle\sum_i \xi_i^\mu s_i$

Energy: $E(\boldsymbol{s}) = -\dfrac{1}{2} \displaystyle\sum_{i,j} J_{ij} s_i s_j$

Capacity (random dense patterns): $P_{\max}/N \approx 0.138$